

Disproof of Conjecture 00000001022: $\min(k, n - k) \cdot \lceil \log_2 q \rceil$ is not a lower bound on trellis complexity

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2026-10-04

Abstract

Conjecture 00000001022 asserts that, beyond the MacWilliams identities for support weights, every $[n, k]_q$ linear code has trellis complexity at least $\min(k, n - k) \cdot \lceil \log_2 q \rceil$. This is Wolf’s classical *upper* bound turned into a lower bound, and it is false. The binary $[10, 5]$ code $C = \{00, 11\}^5$ (the direct sum of five repetition codes of length 2) has a trellis with layer sizes $1, 2, 1, 2, \dots, 1$. In it, the maximum number of states is 2, the maximum number of edges per section is 2, there are 16 vertices and 20 edges in total, and the Viterbi algorithm needs 25 operations. Every one of these measures has $\log_2$ below the claimed bound 5. Its minimal (BCJR) trellis has state profile $s_i = 0, 1, 0, 1, \dots, 0$, so $s_{\max} = 1 < 5$. More generally, $\{00, 11\}^m$ is a $[2m, m]_2$ code with $s_{\max} = 1$ against a claimed bound of m , so the gap is unbounded. The refutation covers state, edge, total-vertex, total-edge and Viterbi complexity on a logarithmic scale, as well as raw maximum state and edge counts. It holds for any trellis, for the minimal trellis, and for the minimum over coordinate orders. A separate $[4, 2]$ example also defeats the *worst* coordinate order. Trellises, path labels, the complexity measures, the BCJR profile and the whole family are formalized in Lean 4 (core library only).

1 The conjecture and the clause we refute

The statement reads:

Conjecture: Beyond the MacWilliams identities for support weights of subcode subsets of any $[n, k]_q$ code, the lower bound for trellis complexity is $\min(k, n - k) \cdot \lceil \log_2 q \rceil$ (a trellis lower bound).

The Chinese version says the same. The opening phrase “beyond the MacWilliams identities for support weights” is a preamble. The MacWilliams-type identities for support weight distributions (Kløve, Simonis) are known theorems, and the statement attaches no claim to them that the bound depends on. If the preamble is read as a conjunct “MacW”, the conjecture is $\text{MacW} \wedge (\text{TLB})$, and refuting (TLB) refutes the conjunction, whatever MacW says. The substantive clause is:

(TLB) For every linear $[n, k]_q$ code C , $\text{TC}(C) \geq \min(k, n - k) \cdot \lceil \log_2 q \rceil$,

where TC is “trellis complexity”. The statement does not define TC, so Section 5 goes through every standard meaning. The bound carries the factor $\lceil \log_2 q \rceil$, which is the number of bits per q -ary symbol. So TC is a logarithmic (bit) measure, and for $q = 2$ the factor is 1. Since (TLB) is claimed for all q , it suffices to refute its instance $q = 2$.

Where the bound comes from. Wolf [2] showed that the syndrome trellis of an $[n, k]_q$ code has at most $q^{\min(k, n-k)}$ states at each time. So the state complexity satisfies $s_{\max} \leq \min(k, n-k)$ in q -ary units, that is, at most $\min(k, n-k) \log_2 q$ bits. This is an *upper* bound, valid for every code, and it is attained by many codes (for example MDS codes). As a lower bound it fails badly, as we now show.

2 Trellises and complexity measures

Definition 1. A (binary) trellis of length n consists of vertex layers $V_0, \dots, V_n$ with $|V_0| = |V_n| = 1$ (root and toor) and edge sections $E_1, \dots, E_n$, where each $e \in E_i$ goes from a vertex of V_{i-1} to a vertex of V_i and carries a label in $\mathbb{F}_2$. The label sequence of a root-to-toor path is a word in $\mathbb{F}_2^n$. The trellis *represents* $C \subseteq \mathbb{F}_2^n$ if the set of these label sequences is exactly C .

The standard complexity measures of a trellis T are

- state complexity $s_{\max}(T) = \max_i \log_2 |V_i|$ (“trellis width”);
- edge (branch) complexity $b_{\max}(T) = \max_i \log_2 |E_i|$;
- total complexity $\log_2 |V|$ and $\log_2 |E|$, where $|V| = \sum |V_i|$ and $|E| = \sum |E_i|$;
- Viterbi complexity $\log_2(2|E| - |V| + 1)$. This counts the additions and comparisons of the Viterbi algorithm [5].

The *trellis complexity* of a code C is the minimum of the chosen measure over all trellises representing C . Some authors also minimize over all coordinate orders of C . Alternatively, it is defined as the measure of the *minimal* trellis of C . A linear code has a unique minimal trellis, the BCJR/Forney/Muder trellis [1, 3, 4, 5, 6]. It minimizes every $|V_i|$ and every $|E_i|$ simultaneously, so it minimizes all the measures above. Every reading of $\text{TC}(C)$ is therefore bounded above by the measure of *any one* trellis representing C :

$$\text{TC}(C) \leq \mu(T) \quad \text{for every trellis } T \text{ representing } C. \quad (1)$$

To refute (TLB) it thus suffices to exhibit one code and one trellis for it whose measure lies below the claimed bound.

The minimal trellis. For a linear code C of length n and $0 \leq i \leq n$, let $P_i = \{c \in C : c_j = 0 \ \forall j \geq i\}$ (past subcode) and $F_i = \{c \in C : c_j = 0 \ \forall j < i\}$ (future subcode). The minimal trellis has $|V_i| = |C|/(|P_i| |F_i|)$ states at time i , that is,

$$s_i = k - \dim P_i - \dim F_i \quad (\text{in } q\text{-ary units}) \quad (2)$$

[3, 5, 6]. Equivalently, $s_i = \text{rk } G_{[0,i)} + \text{rk } G_{[i,n)} - k$ for any generator matrix G .

3 The counterexample $\{00, 11\}^5$

Let $R = \{00, 11\} \subseteq \mathbb{F}_2^2$ be the binary repetition code of length 2, and let

$$C = R^5 = R \oplus R \oplus R \oplus R \oplus R = \{(a_1, a_1, a_2, a_2, \dots, a_5, a_5) : a_j \in \mathbb{F}_2\} \subseteq \mathbb{F}_2^{10}.$$

It is generated by the five words $e_{2j} + e_{2j+1}$, $0 \leq j \leq 4$ (coordinates are numbered $0, \dots, 9$). These have disjoint supports, so they are linearly independent and C is a binary $[10, 5]$ code with 32 codewords. The claimed bound is $\min(5, 5) \cdot \lceil \log_2 2 \rceil = 5$.

Theorem 2. Let T be the trellis with layers $|V_{2j}| = 1$ ($V_{2j} = \{0\}$) and $|V_{2j+1}| = 2$ ($V_{2j+1} = \{0, 1\}$), and sections

$$E_{2j+1} = \{0 \xrightarrow{0} 0, 0 \xrightarrow{1} 1\}, \quad E_{2j+2} = \{0 \xrightarrow{0} 0, 1 \xrightarrow{1} 0\} \quad (0 \leq j \leq 4).$$

Then T represents C , and

$$\max_i |V_i| = 2, \quad \max_i |E_i| = 2, \quad |V| = 16, \quad |E| = 20, \quad 2|E| - |V| + 1 = 25.$$

Hence $s_{\max}(T) = b_{\max}(T) = 1$, $\log_2 |V| = 4$, $\log_2 |E| \approx 4.32$ and $\log_2 25 \approx 4.64$. All of these are < 5 .

Proof. A root-to-leaf path passes through the single vertex of V_{2j} at every even time. Between times $2j$ and $2j+2$ there are exactly two sub-paths: $0 \xrightarrow{0} 0 \xrightarrow{0} 0$ and $0 \xrightarrow{1} 1 \xrightarrow{1} 0$, with labels 00 and 11. Indeed, from state 0 of V_{2j+1} the only outgoing edge has label 0, and from state 1 the only one has label 1. The paths are therefore the free concatenations of five such blocks, and their labels are exactly the words $(a_1, a_1, \dots, a_5, a_5)$, which is C . The counts are immediate: the layer sizes are $1, 2, 1, 2, \dots, 2, 1$ (sum $6 + 10 = 16$), and each of the 10 sections has 2 edges (sum 20). $\square$

Proposition 3. The minimal trellis of C has state profile $(s_0, \dots, s_{10}) = (0, 1, 0, 1, \dots, 1, 0)$, so $s_{\max} = 1 < 5$. Its sections have 2 edges each, so $b_{\max} = 1$.

Proof. A codeword vanishes on the coordinates $\geq i$ iff all blocks meeting $\{i, \dots, 9\}$ are 00. For $i = 2j$ this leaves the j blocks before i free, and for $i = 2j+1$ it leaves the j blocks strictly before block j free. Hence $\dim P_{2j} = j$ and $\dim P_{2j+1} = j$. Symmetrically, $\dim F_{2j} = 5 - j$ and $\dim F_{2j+1} = 5 - j - 1$. By (2), $s_{2j} = 5 - j - (5 - j) = 0$ and $s_{2j+1} = 5 - j - (4 - j) = 1$. The trellis of Theorem 2 has these layer sizes, so by uniqueness of the minimal trellis it is the minimal trellis. In particular the minimal trellis has 2 edges per section. (The concrete numbers $|P_i| = 1, 1, 2, 2, \dots, 16, 16, 32$ and $|F_i| = 32, 16, 16, \dots, 2, 2, 1, 1$ are also checked in Lean and in `verify.py`.) $\square$

Corollary 4. (TLB) is false for $q = 2$, and hence false, under each of the readings listed in Section 5(a)–(e).

Proof. By (1), Theorem 2 and Proposition 3, every reading of $\text{TC}(C)$ in (a)–(c) and (e) is at most $\log_2 25 < 5$, and many are equal to 1. For the rounded readings in (d), see the discussion there; the only case that needs the next member $\{00, 11\}^6$ of the family below is rounding up $\log_2 |E|$ and the Viterbi count. $\square$

4 A whole family: $\{00, 11\}^m$

The same construction works for every $m \geq 1$. The code R^m is a $[2m, m]_2$ code, and the trellis with layer sizes $1, 2, 1, 2, \dots, 1$ (the m -fold concatenation of the two-section trellis of R) represents it. The argument is the proof of Theorem 2 with 5 replaced by m . The minimal profile is $0, 1, 0, 1, \dots, 0$ by the proof of Proposition 3. The trellis has

$$\max |V_i| = 2, \quad \max |E_i| = 2, \quad |V| = 3m + 1, \quad |E| = 4m, \quad 2|E| - |V| + 1 = 5m,$$

against a claimed bound of m bits, that is, 2^m in raw counts.

m	bound	$s_{\max}$	$b_{\max}$	$\log_2 V $	$\log_2 E $	$\log_2(5m)$	measures < bound
1	1	1	1	2	2	2.32	none
2	2	1	1	2.81	3	3.32	$s_{\max}, b_{\max}$
3	3	1	1	3.32	3.58	3.91	$s_{\max}, b_{\max}$
4	4	1	1	3.70	4	4.32	$s_{\max}, b_{\max}, V $
5	5	1	1	4	4.32	4.64	all five
$m \geq 5$	m	1	1	$< m$	$< m$	$< m$	all five

For $m \geq 5$ we have $5m < 2^m$ (by induction: $25 < 32$, and $5(m+1) = 5m + 5 < 2^m + 2^m$), and therefore also $3m+1 < 2^m$ and $4m < 2^m$. So every member with $m \geq 2$ refutes (TLB) for state and edge complexity, and every member with $m \geq 5$ refutes it for all five measures. The gap between the claimed bound m and the true $s_{\max} = 1$ is unbounded.

The smallest instance is the $[4, 2]$ code $\{0000, 1100, 0011, 1111\}$ ($m = 2$). Its minimal profile is $0, 1, 0, 1, 0$, so $s_{\max} = 1 < 2 = \min(2, 2)$.

Remark 5 (q -ary analogue). For any q , the code $\{aa : a \in \mathbb{F}_q\}^m$ is a $[2m, m]_q$ code. It has the analogous trellis with layer sizes $1, q, 1, q, \dots, 1$, so $s_{\max} = \log_2 q$ bits, while the claimed bound is $m \lceil \log_2 q \rceil$. For $q = 3$ the bound fails even at $m = 1$ ($\log_2 3 \approx 1.585 < 2$). `verify.py` checks $q = 3, 5$; this remark is not in Lean.

5 Readings covered

- (a) **Measure.** State complexity $s_{\max}$, edge complexity $b_{\max}$, total vertex complexity $\log_2 |V|$, total edge complexity $\log_2 |E|$, and Viterbi complexity $\log_2(2|E| - |V| + 1)$. All five are below 5 for $C = R^5$.
- (b) **Which trellis.** The minimum over all trellises representing C , or the minimal (BCJR) trellis. By (1) one trellis suffices, and in addition the BCJR profile is computed directly (Proposition 3). Our trellis is in fact the minimal one.
- (c) **Coordinate order.** The given order, or the minimum over all $10!$ coordinate permutations (“trellis complexity of the equivalence class”). A minimum over orders is at most the value for the identity order.
- (d) **Units and rounding.** For $q = 2$, bits and q -ary units coincide and $\lceil \log_2 q \rceil = 1$. The values $s_{\max} = b_{\max} = 1$ and $\log_2 |V| = 4$ are integers, so rounding $\log_2$ up or down changes nothing for them. Rounding *down* keeps all five measures below 5. Rounding *up* turns $\log_2 20$ and $\log_2 25$ into exactly 5, so for these two measures R^5 does not refute the rounded-up reading. The next member R^6 does: $\lceil \log_2 19 \rceil = \lceil \log_2 24 \rceil = \lceil \log_2 30 \rceil = 5 < 6$.
- (e) **Raw state and edge counts.** Even without logarithms, $\max |V_i| = \max |E_i| = 2 < 5$.
- (f) **Worst coordinate order.** Some authors might take the *worst* order (“some ordering of C needs $\min(k, n-k)$ bits”). This reading needs another example, because R^5 in the interleaved order $0, 2, 4, 6, 8, 1, 3, 5, 7, 9$ has $s_{\max} = 5$. The $[4, 2]$ code generated by 1000 and 0111 has no zero coordinate, and in every one of its 24 coordinate orders its minimal trellis has $s_{\max} = 1 < 2$. In order to have $s_i = 2$ one needs $P_i = F_i = 0$, so the first coordinate (the only one that sees the generator 1000) would have to lie both before and after time i .

The only escape: unitless total counts. If “trellis complexity” were the raw number $|V|$ or $|E|$ of vertices or edges, compared with $\min(k, n - k) \lceil \log_2 q \rceil$ as a pure number, then R^5 would not be a counterexample ($16, 20 \geq 5$). We do not claim to refute that reading, but it is not a sensible reading of the statement. The bound is stated in bits ($\lceil \log_2 q \rceil$ bits per symbol), and a raw count of vertices or edges is not measured in bits. Moreover, for $|E|$ the raw inequality is trivially true and says nothing about trellis structure. Every $|E_i| \geq 1$ and $x \geq 1 + \log_2 x$ for every positive integer x . A q -ary code has q^k paths, so $\prod_i |E_i| \geq q^k$. Hence $|E| \geq n + \sum_i \log_2 |E_i| \geq n + k \log_2 q \geq k(\log_2 q + 1) \geq \min(k, n - k) \lceil \log_2 q \rceil$.

Non-vacuity. The clause is not refuted for a trivial reason. The interleaved $[4, 2]$ code $\{0000, 1010, 0101, 1111\}$ has minimal profile $0, 1, 2, 1, 0$, so it *attains* the bound 2, and Wolf’s bound is tight there. For $R = \{00, 11\}$, *every* trellis needs 2 states at time 1: if the paths for 00 and 11 met in a common middle state, then 01 would also be a path label. So (TLB) holds for R over all trellises. An exhaustive scan (`verify.py`) of all $2 + 5 + 16 + 67 + 374 + 2825$ binary linear codes of length $n \leq 6$ confirms Wolf’s upper bound $s_{\max} \leq \min(k, n - k)$ for every one of them, and finds that 0, 2, 6, 37, 200, 1785 of them violate the claimed lower bound.

A different statement. One might read the conjecture as saying that $\min(k, n - k) \cdot \lceil \log_2 q \rceil$ is the best *universal upper* bound, that is, Wolf’s bound. That is a different (known) statement, not a lower bound “for any $[n, k]_q$ code”, and we do not address it.

6 Formal verification (Lean 4)

The directory `lean4/` contains a self-contained Lean 4.19.0 project, using the core library only. All objects are defined from scratch:

- words are `List Bool`; `span n gens` lists the 2^k $\mathbb{F}_2$ -combinations of the generators; `IsGen n k gens` says that there are k generators of length n that are linearly independent (all 2^k combinations are distinct);
- `Trellis` consists of a list of layer sizes and a list of sections of labelled edges (from, bit, to). `WF` requires one root, one toor, edge endpoints in the adjacent layers and no repeated edge. `IsPath` defines a root-to-toor path with a given label sequence. `Represents T n gens` says that the path labels are exactly the codewords. `labels` enumerates the paths, and `mem_labels` proves that the enumeration is correct;
- the measures `maxStates`, `maxEdges`, `numVertices`, `numEdges` and `viterbiOps`. The clause “ $\log_2 X \geq B$ ” is stated as $2^B \leq X$, which is equivalent for the real logarithm. `CeilLog2 q e` characterizes $e = \lceil \log_2 q \rceil$ (and `ceilLog2_unique` proves uniqueness);
- `pastSub`, `futSub` and `minProfile`, the BCJR profile $|C|/(|P_i||F_i|)$; `permW` and `IsPermOf` for coordinate orders.

The clause is `Clause  $\mu$` : for all n, k , generators, e with `IsGen n k gens` and `CeilLog2 2 e`, and every well-formed trellis T representing the code, $2^{\min(k, n - k) \cdot e} \leq \mu(T)$. The main results are:

- `ex10_isGen`, `ex10_wf`, `ex10_represents`, `ex10_numbers` (the values 2, 2, 16, 20, 25), `ex10_minProfile` (1, 2, 1, ..., 1) and `ex10_pastFut`;

- `clause_false` : $\forall \mu, \neg \text{Clause } \mu$ and `conjecture_00000001022_false` : $\forall \text{MacW } \mu, \neg \text{Conjecture MacW } \mu$;
- `clauseMin_false`, `clausePerm_false` and `clauseMinPerm_false`, for the minimal-trellis and coordinate-order readings;
- the family, for *all* m : `family_isGen`, `family_wf`, `family_represents` and `family_numbers` (≤ 2 , ≤ 2 , $3m + 1$, $4m$); then `family_viterbi` ($5m$), `family_gap_states` ($m \geq 2$), `family_gap_all` ($m \geq 5$), `clause_false_family` and `clause_false_family_all`; `family_minProfile_small` gives the BCJR maximum 2 for $m \leq 6$;
- `worstOrder_false`, the worst-order reading, via the $[4, 2]$ code $\langle 1000, 0111 \rangle$ and all orders;
- non-vacuity: `rep2_clause_holds` (every trellis for $\{00, 11\}$ has ≥ 2 states) and `inter4_meets_bound`.

The project contains no `sorry`, no `native_decide` and no added axioms. `#print axioms` reports at most `propext`, `Classical.choice` and `Quot.sound`.

Limitations. The Lean formalization is binary ($q = 2$), which suffices to refute a statement made for all q . The theorem that the minimal trellis has $|C|/(|P_i||F_i|)$ states (equation (2)) is cited, not formalized. Lean computes this quantity, and, independently, the explicit trellis already gives the upper bound (1). The BCJR profile of the family for general m , the q -ary remark and the raw- $|E|$ inequality are proved in this report only (the first two are checked numerically in `verify.py`).

7 Independent check

`verify.py` (Python 3 standard library) uses different methods. It computes the BCJR profile from ranks of generator submatrices (Gaussian elimination mod p) and cross-checks it by counting. It builds the minimal trellis explicitly as the pruned syndrome trellis from a parity-check matrix, and checks its layer sizes and its path labels. It also expands the explicit trellis forward, covers the family for $m \leq 8$, $q = 3, 5$ and all coordinate orders of the $[4, 2]$ examples, and scans all binary linear codes of length ≤ 6 .

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